DEWATERING IN CLOSED PITS MINING USING SOLAR POWER

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Abstract— The development of a solar-powered dewatering system for mining operations represents a significant advancement in sustainable water management technology. This project aims to design an efficient dewatering setup capable of removing accumulated water from mine pits while utilizing renewable solar energy as its primary power source. The challenges inherent in this endeavour include maintaining continuous operation under varying sunlight conditions, optimizing energy storage, and integrating mechanical and electrical components into a reliable system. To address these challenges, the project incorporates best practices such as system optimization, modular design, and the use of energy-efficient pumping mechanisms. The successful implementation of this solar-based dewatering system can lead to reduced operational costs, minimized environmental impact, and improved sustainability in mining activities. As the project progresses, the focus on enhancing system efficiency and operational reliability remains paramount. The commitment to continuous innovation and adaptation of emerging solar technologies will be essential for the dewatering system to meet the evolving energy and environmental demands of modern mining operations.

Keywords — Solar Panel, Dewatering, Mining, Renewable Energy, Sustainability, Water extraction

I. INTRODUCTION

In the modern industrial landscape, efficient and sustainable water management has become a crucial requirement, especially in mining operations where closed pits frequently accumulate large volumes of water. Dewatering plays an essential role in maintaining operational safety, preventing equipment damage, and ensuring uninterrupted mining activities. Traditional dewatering methods rely heavily on diesel engines or grid-based electricity, which lead to high operational costs, fuel dependency, and adverse environmental impacts. To overcome these challenges, this project focuses on the design and development of a solar-powered dewatering system specifically for closed pit mining applications

Solar technology has advanced significantly in recent years, making it a reliable and cost-effective source of renewable energy. Its integration into dewatering systems offers advantages such as reduced carbon emissions, lower maintenance requirements, and the ability to operate in remote locations without grid connectivity. The motivation behind this project stems from the increasing demand for eco-friendly alternatives in mining operations and the need for continuous water removal even under varying environmental conditions. By utilizing solar panels, energy storage units, and sensor-based automation, the proposed system aims to deliver consistent and efficient dewatering performance.

The primary objective of this project is to design an automated dewatering system that uses photovoltaic energy to power the pump and control unit.

Scope and Significance:

The scope of this work involves the integration of solar energy modules, sensor-based water-level monitoring, relay-controlled pumping mechanisms, and a microcontroller platform to implement an intelligent dewatering setup. The significance lies in demonstrating a practical, low-cost, and sustainable solution capable of functioning effectively in mining environments where conventional power sources are either costly or unavailable.

Developing a solar-powered dewatering system presents several challenges, including fluctuating solar irradiance, energy storage limitations, hardware integration, and real-time sensing accuracy. The project addresses these challenges by selecting efficient components, optimizing system design, and implementing automated safety controls to improve performance and reliability.

The methodology adopted in this project includes:

Understanding the principles of dewatering and solar power generation.

Selecting appropriate hardware components such as solar panels, sensors, relay module, and pump.

Designing algorithms for sensor data processing and automated pump control.

Integrating hardware and software elements to create a functional solar-powered system.

Testing the system under varying conditions to evaluate operational accuracy and stability.

This report is organized into multiple sections to provide a complete overview of the system development process. It includes the literature review, system design, hardware and software implementation, experimental results, discussion, conclusion, and recommendations for future improvements.

Overall, this project aims to contribute to sustainable mining practices by demonstrating an efficient, environmentally, friendly and autonomous solar power dewatering system.

II. LITERATURE REVIEW

Solar-based pumping and water-management systems have been widely studied, with a strong focus on energy efficiency, automation, and suitability for remote locations. A solar-powered dewatering approach was presented in **[1]**, where solar panels were used to operate pumps for removing water from mining sites. The study demonstrated that solar dewatering provides an environmentally friendly and sustainable alternative to conventional fossil-fuel-driven systems.

A photovoltaic-operated water pumping design was discussed in **[2]**, highlighting that PV-based pumping offers a reliable and cost-effective solution for remote regions where grid electricity is limited or unavailable. The primary objective of the system was to reduce dependency on traditional power sources.

Automation of water level monitoring was explored in **[3]**, where ultrasonic sensors combined with a microcontroller were used to develop an automated water-level sensing and pump-control system. This approach enabled real-time level detection and reduced risks of overflow. A similar sensor-based analysis was provided in **[5]**, which evaluated the accuracy and detection performance of ultrasonic sensors, confirming their suitability for real-time water-level monitoring applications.

A broader overview of solar-energy-based pumping and water management was presented in **[4]**, emphasizing the scalability and environmental benefits of solar power. A more advanced solar-powered pumping and industrial control methodology was introduced in **[7]**, where the integration of automated control circuitry improved water regulation and system efficiency.

A detailed implementation of automated pump operation was described in **[8]**, where sensors and Arduino-based logic were used to design a fully automated water-pumping mechanism.

This approach removed the need for manual intervention and ensured more intelligent regulation of water flow.

A comprehensive review of solar water pumping technologies was provided in **[9]**, summarizing system configurations, performance challenges, economic considerations, and overall applicability. The review concluded that solar pumping systems, despite higher initial costs, offer long-term sustainability and require significantly less maintenance compared to conventional pumping systems.

A visual demonstration of solar pump functioning was presented in **[6]**, showing the practical operation of a solar panel–driven motor and water-lifting mechanism. This demonstration supports the fundamental concepts discussed in previous studies.

Overall, the reviewed literature indicates that solar-powered pumping systems are highly effective for both remote and industrial applications. Automation through sensor integration further enhances system reliability and water-level control. However, many existing studies do not offer a fully integrated, low-cost system combining solar pumping with advanced sensing and intelligent automated control, creating a clear opportunity for further improvement in system design.

III. PROBLEM STATEMENT

Closed pit mines frequently face excessive water accumulation caused by rainfall, seepage, and groundwater inflow. Traditional pumping systems require diesel or grid electricity, resulting in high operational costs and environmental impact. Therefore, an automated, renewable-energy-based solution is required to monitor water levels and operate pumping mechanisms reliably. This project aims to design and implement a solar-powered automated dewatering system using sensors such as ultrasonic, DHT11, and float switch, along with a relay-based pump control unit. The system must function efficiently under varying solar conditions and ensure uninterrupted dewatering for maintaining safe mining operations, preventing equipment damage, and enabling continuous water removal even in remote or off-grid environments.

IV. SYSTEM DESIGN

A. Methodology of Model Development

The model development begins with integrating sensors, including the ultrasonic sensor for water level measurement, DHT11 for environmental monitoring, and float switch for safety water-level detection. These sensors are interfaced with the Arduino UNO, which serves as the central processing unit. The Arduino processes sensor data and controls the relay module that operates the dewatering pump. The pump is powered using a photovoltaic (PV) solar panel, supported by a battery system for energy storage. The processed sensor *data is* displayed on an LCD interface, allowing real-time monitoring of water levels and environmental conditions. This system ensures automatic pump activation when water exceeds a defined level and deactivation once the pit is drained to a safe level.

B. Hardware Module Design

The hardware design includes components such as Arduino UNO, ultrasonic sensor, DHT11 sensor, float switch, relay module, solar panel, battery, and pump. The ultrasonic sensor measures pit water levels without physical contact. The float switch acts as a backup safety mechanism. The relay module switches the pump ON/OFF based on Arduino signals. The solar panel supplies power to the system, while the battery ensures continuous operation during low-sunlight conditions. All components are mounted on a breadboard and connected via jumper wires for ease of testing and implementation.

C. Block Diagram

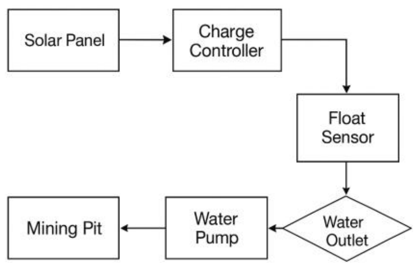

Fig.1 Block diagram for solar-powered dewatering system

V. HARDWARE DESCRIPTION

A. Ultrasonic Sensor

An ultrasonic sensor measures water level in the pit by transmitting high-frequency sound waves and calculating the time taken for echoes to return from the water surface. This enables accurate, non-contact water-level monitoring, essential for muddy mining environments.

Fig.2 Ultrasonic Sensor

B. DHT11 Sensor

The DHT11 measures temperature and humidity around the mining pit. These atmospheric factors influence evaporation rate, system performance, and solar power generation efficiency. The DHT11 provides stable digital output and ensures environmental monitoring for optimized operation.

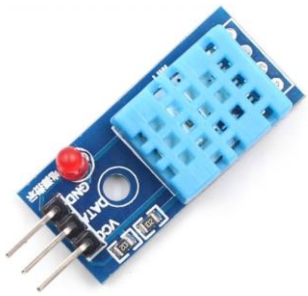

Fig.3 DHT11 Sensor

C. Float Switch

The float switch acts as a mechanical water-level safety controller. It rises and falls with water level; when water exceeds a preset height, it triggers pump activation. When water goes below a certain point, it switches the pump OFF. It prevents dry-running and provides redundancy to the ultrasonic sensor.

Fig.4 Float switch

D. Relay Module

A relay module serves as an electrical switch for controlling the dewatering pump. It receives low-voltage signals from Arduino and switches the high-voltage pump circuit ON/OFF, enabling automated water removal.

Fig.5 Relay Module

E. Arduino UNO

Arduino UNO is the central microcontroller that reads sensor data, processes conditions, and sends commands to the relay to operate the pump. It coordinates all sensing and control operations in the dewatering system. It continuously monitors the water level and decides when pumping is required. It ensures the pump runs only under safe and predefined conditions, preventing overflow or dry run. It acts as the main decision-making unit, connecting all hardware components into one controlled automation system.

Fig.6 Arduino UNO Board

VI. HARDWARE WORKING

The solar-powered dewatering system operates by continuously monitoring the water level using the ultrasonic sensor. When the water level crosses the predefined threshold, the Arduino activates the pump through the relay module. The float switch serves as a backup level detector to ensure pumping stops in case water reaches a minimum safe level. The DHT11 sensor monitors environmental conditions to support system performance analysis. The LCD display shows real-time water level and sensor readings. The entire setup runs on renewable solar energy, stored in a battery for night-time or cloudy conditions.

VII. SOFTWARE OVERVIEW

The Arduino IDE is used to program the microcontroller, write sensor-reading functions, and control relay switching logic. The software reads ultrasonic and float-switch inputs, executes conditional logic, updates the LCD display, and activates the pump accordingly.

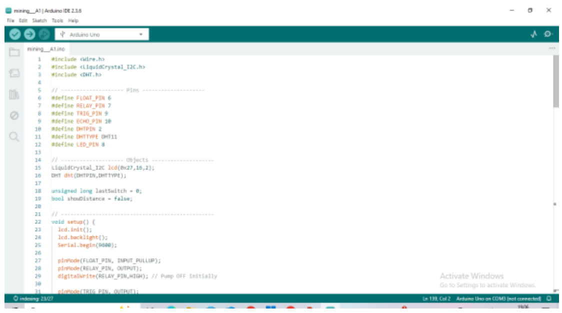

Fig.7 Arduino IDE

VIII. SOFTWARE IMPLEMENTATION

The program works by using the ultrasonic sensor to measure the water level. It reads the echo timing and converts it into distance using the usual timing-based formula. At the same time, it also collects temperature and humidity values from the DHT11 sensor and checks the float switch to know when the water is too high or too low. Using these inputs, the code decides when the pump should turn ON or OFF through the relay module so that the water level stays under control. The LCD display continuously shows the water level, sensor readings, and the status of the system, which makes it easy for the operator to understand what is happening and confirm that everything is running properly even when conditions change. Overall, the program ensures smooth, reliable, and intelligent control of the entire dewatering process.

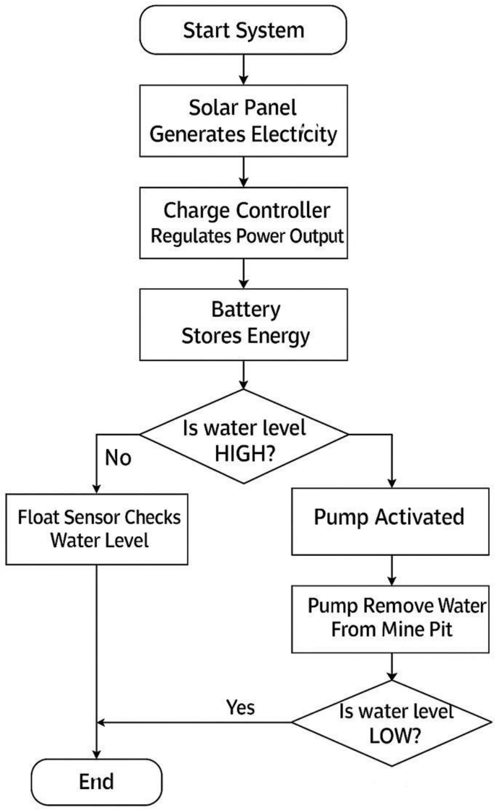

Fig.8 Flow chart

IX. EXECUTION OF MODEL

The final hardware model integrates all sensors and the pump system through the relay module, with the entire setup powered by a solar panel and battery backup. The system continuously monitors the water level, and whenever the sensor readings indicate that the water has reached a high point, the pump turns on automatically. Once the pit is drained to a safe level, the pump switches off without requiring any manual control. This setup ensures smooth and reliable operation, even in remote areas, and shows how the complete system works in real conditions.

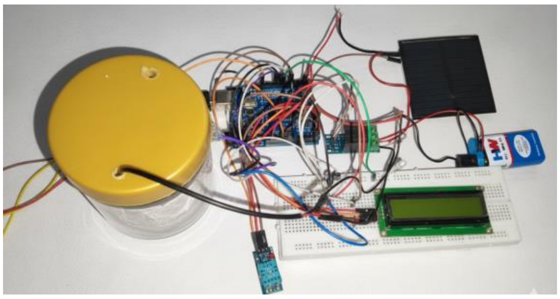

Fig.9 Working Model

X. ADVANTAGES

- Solar-powered dewatering reduces overall operational costs by removing the need for diesel or grid electricity.
- The system eliminates fuel dependency, making it suitable for remote mining areas where power supply is limited.
- Carbon emissions are greatly reduced since the system runs on clean and renewable solar energy.
- With a battery backup, the system can operate continuously, ensuring 24/7 autonomous dewatering.
- Sensors improve measurement accuracy and make the system more reliable in detecting actual water levels.
- Automated sensing and control enhance safety by preventing overflow and equipment damage in pit areas.

XI. APPLICATION

Solar-based automated dewatering systems can be used in a wide range of areas where water removal is necessary and reliable electricity is not available. They are highly useful in mining operations for managing accumulated pit water, and in construction sites where continuous dewatering is required to maintain safe working conditions. These systems are also suitable for agricultural fields that need controlled water removal, as well as in stormwater management setups to prevent flooding during heavy rainfall. Additionally, they are effective in remote locations for groundwater extraction, making them a practical choice wherever power supply is limited or inconsistent.

XII. CONCLUSION

This project provides a practical and effective solution for automated dewatering in closed pit mines using solar energy. The integration of the ultrasonic sensor, float switch, DHT11 sensor, and relay-controlled pump demonstrates how basic sensing and control components can work together to maintain safe water levels without manual involvement. By relying on solar power and battery backup, the system reduces operational costs, minimizes environmental impact, and ensures reliable performance even in remote locations. Overall, the project achieves its objective of creating a sustainable, efficient, and eco-friendly dewatering system suitable for real-world mining applications.

XIII. REFERENCE

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